Nader Khalil

AI Engineer · Agentic Systems & Automation

nader_khalil@outlook.com | +971 56 123 8877 | Abu Dhabi, UAE
linkedin.com/in/naderkhz/ | naderkhalil.dev



PROFESSIONAL SUMMARY

AI Engineer with a B.Sc. in Software Engineering and Unix systems training from 42 Abu Dhabi, designing and deploying production Generative AI applications, RAG pipelines, and agent orchestration workflows. Proficient in vector search (pgvector, Pinecone/FAISS), prompt engineering, automated AI evaluation suites (prompt injection defense, citation grounding), and inference optimization (93% prompt caching). Experienced in mentoring UAE youth and engineering teams on agentic AI for ADEK. Bilingual in Arabic (Native) and English (Professional), based in Abu Dhabi.

TECHNICAL SKILLS

Generative AI & LLMs: RAG architectures, Multi-Agent Orchestration, Tool Calling, MCP, n8n Automation, Prompt Caching, LangChain, Vercel AI SDK, GPT, Llama

AI Safety & Evals: Automated prompt injection testing, scope gating, structured output parsing (Zod, Pydantic), hallucination reduction, citation grounding

MLOps, Cloud & CI/CD: Docker, GitHub Actions CI/CD, Azure AI / Cloud platforms, automated test suites (766+ tests in CI), Playwright, Vitest, PyTest

Vector DBs & Retrieval: Vector embeddings, semantic search, pgvector, Pinecone, FAISS, PostgreSQL, MariaDB, SQL schema design, knowledge base ingestion

Software & API Engineering: RESTful APIs, Python (FastAPI), TypeScript, Next.js, React 19, layered architecture, Auth.js, RBAC, web service integration

Languages & Core: Python, TypeScript, JavaScript, SQL, C, C++, Bash, Linux system internals

TECHNICAL PROJECTS

AI Booking Platform, *Production GenAI & Agentic System* 2026
Live build: booking.naderkhalil.dev · Case study: naderkhalil.dev/case-study.html

- Architected an end-to-end RAG pipeline querying admin-managed docs with vector embeddings and role-filtered SQL retrieval (pgvector); answers include verifiable source citations and enforce conversational context isolation.
- Designed agent orchestration with autonomous tool calling and Human-in-the-Loop review: the model proposes calendar actions as structured cards executed only upon explicit user confirmation.
- Built an automated evaluation suite with 120 live checks gating prompt injections, out-of-scope queries, and data leaks; achieved 93% cached input inference efficiency and verified concurrency with 766 tests in CI.

AI Appointment Scheduler, *LangChain Conversational Agent* 2025
Live build: appointment.naderkhalil.dev

- Built a conversational scheduling agent using Python (FastAPI), LangChain, and Next.js, parsing natural-language scheduling requests into structured API payloads for calendar availability checks and slot reservations.

Student-Faculty Services v2, *Higher-Ed Service Platform* 2026
Live build: student-faculty-services-v2.infinityfree.io · Case study: naderkhalil.dev/case-study-v2.html

- Architected a high-reliability educational services API (46 endpoints) in PHP 8.4 and React 19; eliminated double-bookings via database row locks (proven in 8-student race tests); verified by 500+ tests with 100/100 accessibility.

PROFESSIONAL EXPERIENCE

AI Trainer & Mentor, *Department of Education and Knowledge (ADEK)* 07/2026 – 08/2026

- Delivered technical instruction on agentic workflows, multi-agent orchestration (AG2), and autonomous tool calling in n8n for the accredited AgenticVerse program; mentored teams building AI prototypes for ADNOC, FANR, and Khalifa Fund.
- Guided participants through API schema validation, RAG grounding, safety gates (prompt injection defense, human review), and debugging in Python and TypeScript.

QA & Software Tester, *42 Abu Dhabi* 01/2026

- Conducted functional and UI testing on internal academic platforms using black-box scenarios; authored reproduction reports with logs and steps to help developers isolate defects quickly.
- Tested features across browsers and viewports; re-tested defect fixes across user flows to ensure zero regressions before release.

IT Support & Technical Staff, *Ticketmaster* 12/2025

- Provisioned and deployed entry ticketing hardware and access scanners for a high-volume international event; maintained 100% operational uptime through real-time hardware health checks and rapid on-site incident resolution.

Software Engineering Intern, *Timex IT Consulting* 01/2024 – 04/2024

- Developed C# MinimalAPI CRUD endpoints with request validation and standardized HTTP response handling; verified functional defect fixes on client accounting software before production release.

EDUCATION & CERTIFICATIONS

B.Sc. in Software Engineering, *Al Ain University*
Computer Science Diploma, *42 Abu Dhabi*
MOEI × 42 Abu Dhabi AgentEra Hackathon, *Honoree*

CyberQ CTF, Top 10 Finalist, *UAE Cybersecurity Council*
Hack The Box University CTF, *Top 12% (119/1,014 teams)*
RecognAIze Hackathon, *AI on Watch with Datamind*

Languages: Arabic (Native), English (Professional Proficiency)